

PAPER_1365 — Olfactory Shape vs Vibrational Hybrid

Framework: UQFF — Star-Magic v5.27+ | **Tier:** G Bio/Complex (CLOSED) **Calculator:**
calculate_paradox({"paradox": "olfactory_vibrational"})

Master Expression

Vibrational ($\omega_{\text{SCm phonon}}$) + shape ($\Phi_{\text{res geometry}}$)

Match: RESOLVED

Interpretation

Both Turin vibrational and Lock-and-Key shape theories correct — UQFF unifies via THz phonon (vibrational) $\times$ Φ_{res} geometric matching (shape).

UQFF Locked Primitives

- $D_{\text{phys}} = 4$, $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$, $SO_5 = 10$, $A_5 = 60$, $N_{\text{CH}} = 9$
- $K_{\text{MEX}} = 25/12$, $\beta_i = 0.6029$, $\Phi_{\text{res}} = 0.84$, $\Lambda = 0.00729735$, $F_{\text{TRZ}} = 0.1$

NOT REPLACEMENT

UQFF derives via integer-primitive identity. Zero fit parameters.

Copyright — Daniel T. Murphy, daniel.murphy00@gmail.com, Youngstown, OH.